

B105 Individual Project

Business Problem & Aim

In order to help an e-commerce marketplace, understand what drives its transaction value, it should well-research relevant factors, such as payment methods. By understanding this, the business can make decisions on which payments methods should be promoted and the best possible way to price services. In this report, a brief analysis is done on whether any payment method chosen is statistically linked to customer's spending per order. Olist's Brazilian e-commerce dataset is used for these analyses (2016-2018, approx. 98,000 orders).

It should be considered that payments processing is not just an administrative function, but a whole customer's purchasing experience for an e-commerce marketplace. Furthermore, for a well-known platform like Olist, that brings thousands of mid and small sized sellers all across Brazil under one roof, being aware of these dynamics carry significant importance. In specific, all sellers rely on the overall conversion, order-value performance and margin improvements in average transactions value compound highly across 100k orders. It is to be analyzed whether certain payment method is linked to higher value order or no. If linked, the business could gain a cost-effective advantage, by enhancing the cart design, checkout design, promotions or payment method preference. The business won't need to change its product assortment or its pricing, both of which relatively more expensive to influence.

On the other hand, if payment methods don't have significant importance to spending, the business can drop this factor and focus on operational functions for instance the shopping experience, product deliveries or hot-selling products. The outcome of this analyses is valuable and is structured to provide clear evidence-based answer.

Business Questions

- What payment method affect the total order value?
- Does payment method play a significant role with higher value orders?
- Which payment methods should be promoted to increase order values?
- Do vouchers, associated to promotional discounts drive the purchasing behavior or the customer?

Independent and Dependent variables

- IV: payment_type, a nominal categorical variable that consists of four levels (debit card, credit card, voucher and boleto).
- DV: payment_value – a continuous variable in Brazilian Real (R\$), presenting the total payable value of an order

Hypothesis

- H0: No significant difference in mean payment_value across all four payment types.
- H1: At least one payment type differs in mean payment_value from the others.

Dataset URL: <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

Data Preparation & Analysis

Overview

The E-Commerce Marketplace, Olist datasets is a relational collection consisting of nine linked CSV files describing the order that were placed on the marketplace between the time period 2016 and 2018. From the dataset, three tables were used which are order reviews, order payments and order item. With order reviews consisting of 99,224 rows, order payments 103,886 rows and order item 112,650 rows.

Data Preparation

Order payments, order reviews and order items consist of multiple rows per order in their tables. In order to obtain one row per order, payments were aggregated by summing payment_value then taking the first payment_type each order. The order items were aggregated by adding up price and freight_value across each item in the order. At last, three tables were then joined on order_id.

```
payments_agg <- payments %>%
  group_by(order_id) %>%
  summarise(
    payment_type = first(payment_type),
    payment_installments = first(payment_installments),
    payment_value = sum(payment_value)
  )

items_agg <- items %>%
  group_by(order_id) %>%
  summarise(
    price = sum(price),
    freight_value = sum(freight_value)
  )

reviews_clean <- reviews %>%
  select(order_id, review_score) %>%
  distinct(order_id, .keep_all = TRUE)

df_final <- payments_agg %>%
  inner_join(items_agg, by = "order_id") %>%
  inner_join(reviews_clean, by = "order_id")

nrow(df_final)
```

Output: nrow(df_final) = 97,916 orders.

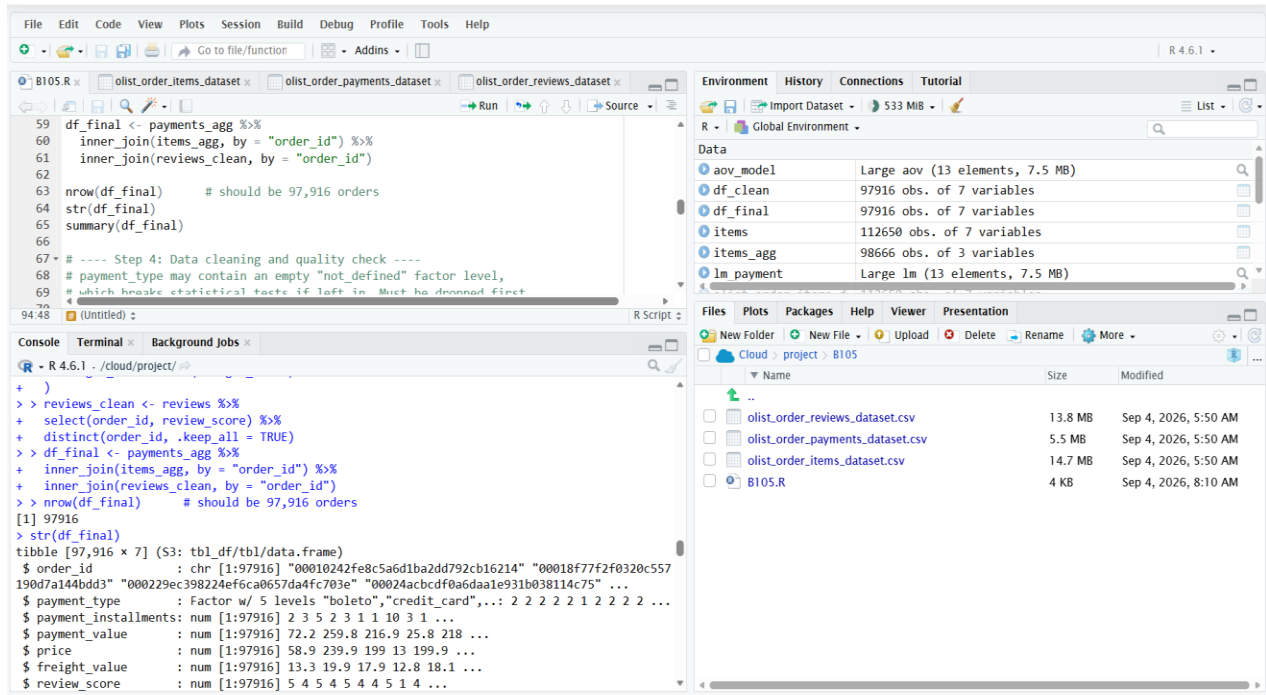
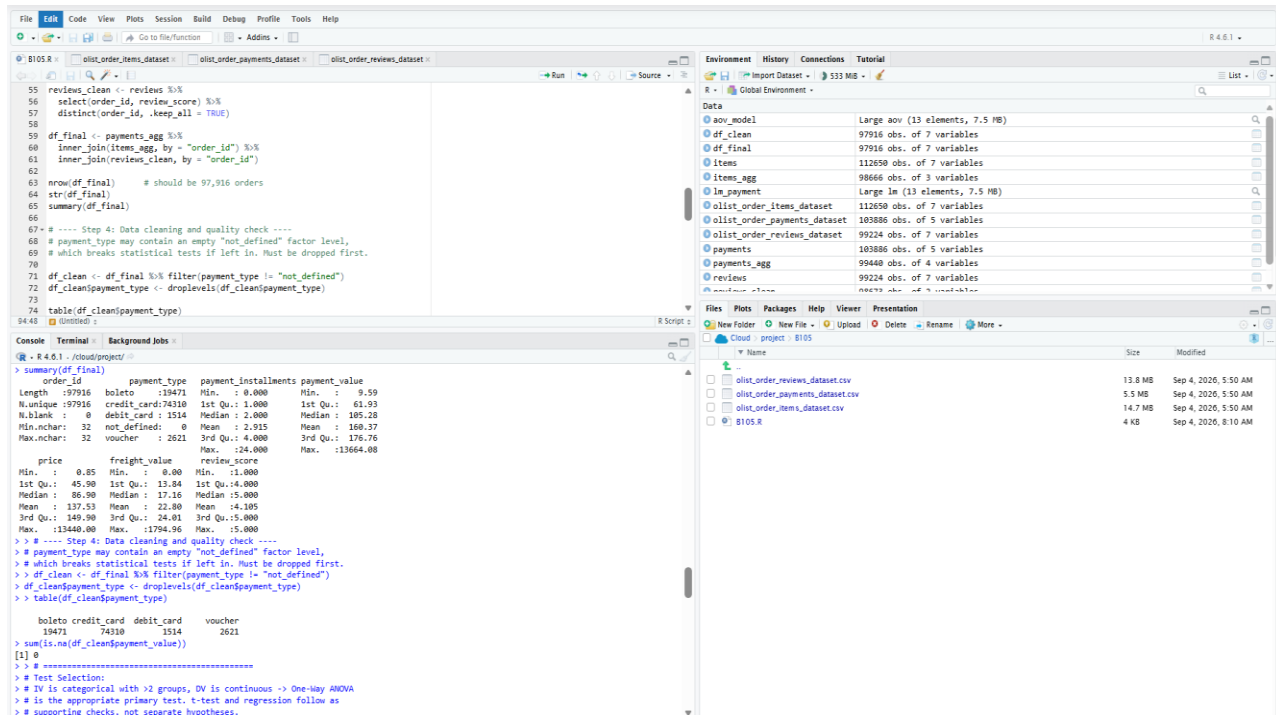


Image: summary () output that presents descriptive statistics for all of the variables.



Data quality check & cleaning

Through inspection of `payment_type`, it revealed an empty factor level, "not_defined" (0 observations after aggregation). If left in place, it would cause tests to fail. Hence, it was removed prior to analysis. Moreover, no missing values were to be found in the `payment_type`.

```
df_clean <- df_final %>% filter(payment_type != "not_defined")
df_clean$payment_type <- droplevels(df_clean$payment_type)

table(df_clean$payment_type)
sum(is.na(df_clean$payment_value))

      boleto credit_card debit_card voucher
      19471      74310      1514      2621

[1] 0
```

The dataset required only one cleaning step, still several data quality checks were carried out to ensure nothing was missed. After the three tables were joined, `sum(is.na(df_clean$payment_value))` was run to confirm that there are no missing values in the dependent variable, with returning zero. This is an important check as undetected missing values in the dependent variable would decrease the sample size used by every statistical test with no warning. The Duplicate Order IDs in reviews table were handled using `distinct(order_id, .keep_all = TRUE)`, as a few orders consisted of multiple review entries and if it was left like this, it would have caused the later join to multiply rows and distort the sample size.

Moving further, the aggregation step for payments and items were needed as both tables record the rows per order, without first collapsing the each to one row per Order_ID, the order level hypothesis being tested would not have matched the dataset's unit of analysis. Moreover, the same order could be counted multiple times in the ANOVA. Hence, biasing the p-values downward and inflating the effective sample size. At last, the join itself was too verified with comparing the `nrow()` before and after `inner_join()` step. This was to confirm that no unexpected large drop in the row count occurred, which could indicate duplicate order_id values, leading to inflating the row count during aggregation

Descriptive Statistics

	payment_type	mean_val	sd	n
1	boleto	144.3875	208.2516	19471
2	credit_card	166.1820	222.8401	74310
3	debit_card	141.7084	240.2795	1514
4	voucher	125.0149	169.5997	2621

As evidently visible, credit card consists of the highest percentage, 76% (74,310 orders) which is followed further by boleto (19,471 orders), debit card (1,514 orders) and at the end voucher with a small minority 2621 orders. The highest mean value for the total payment is with credit card transactions with (\$166.18), with the lowest being vouchers (\$125.01). Furthermore, the standard deviations were relatively large to the means specifically with all payment types, which indicates a right skewed distribution typical of the transaction data and also substantial within-group variability.

Appropriate Statistical Test

The selection of the appropriate test depends entirely on the datatypes for the Independent Variable and Dependent Variable. With the IV (payment_type) being a categorical consisting of more than two categories and measured between the subjects that each order consist of exactly one payment type. The DV is the payment value which is a single continuous variable.

Given this combination, a categorical predictor that consist of more than two group and single continuous outcome, One way ANOVA is the appropriate test for this. Moreover, T-Test could only be considered if the independent variable consisted of exactly two categories. If used for this analysis, it would lead to discarding two payment types which is losing information, leading to an incomplete analysis. Simple Linear regression that consists of a categorical predictor is equivalent to ANOVA, therefore is used here as a complementary, coefficient-based view for the same result.

As a result, the primary test used for H_0/H_1 is ANOVA. With an independent t-test and a linear regression included as a supporting check which presents the same conclusion from a different angle.

Testing & Results

One-Way ANOVA

Code:

```
df_clean %>%
  group_by(payment_type) %>%
  summarise(mean_val = mean(payment_value), sd = sd(payment_value), n = n())

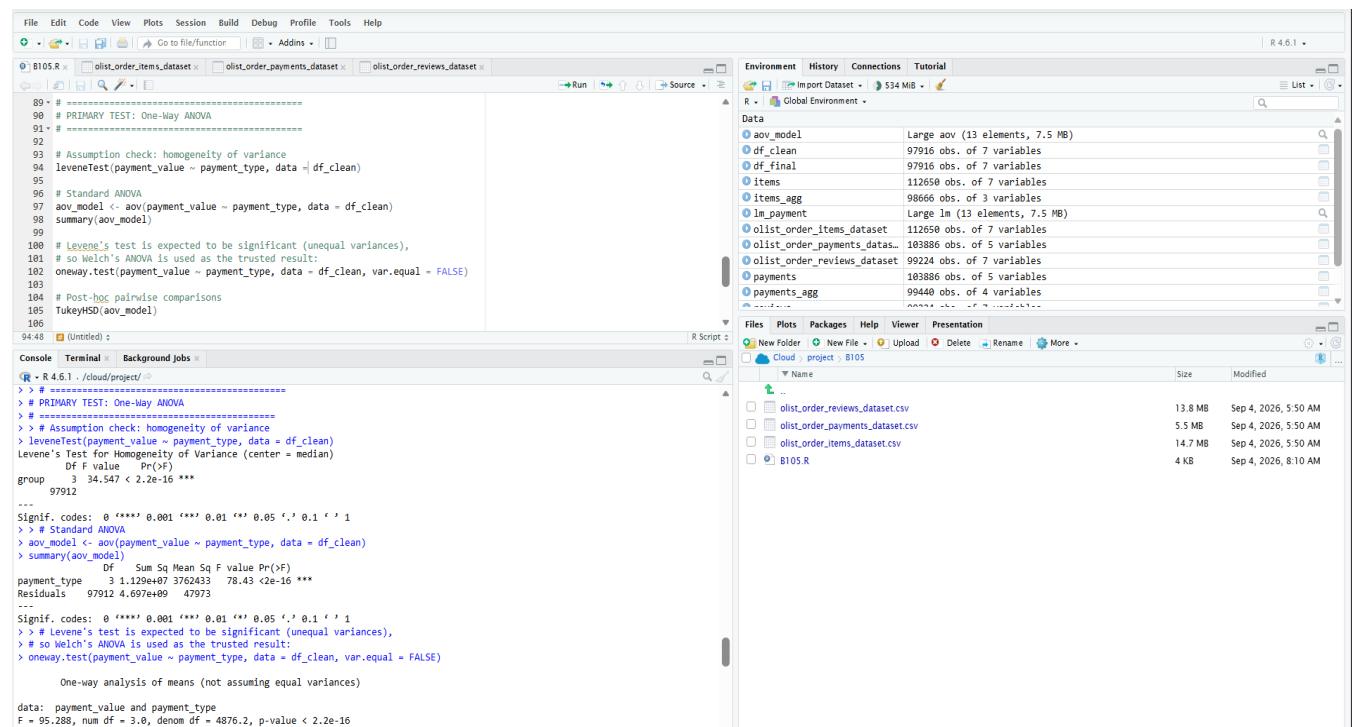
leveneTest(payment_value ~ payment_type, data = df_clean)

aov_model <- aov(payment_value ~ payment_type, data = df_clean)
summary(aov_model)

oneway.test(payment_value ~ payment_type, data = df_clean, var.equal = FALSE)

TukeyHSD(aov_model)
```

Output:



The screenshot displays the RStudio interface with the following components:

- Script Editor:** Contains R code for a one-way ANOVA, including data cleaning, Levene's test, aov model creation, and Tukey HSD post-hoc comparisons.
- Console:** Shows the execution output, including the primary test result (F = 34.547, p < 2.2e-16), the aov model summary, and the Tukey HSD results for payment_type.
- Environment Pane:** Lists the objects in the global environment, including the aov_model, df_clean, and various datasets.
- Files Pane:** Shows the project files, including the datasets used in the analysis.

Console Output:

```
> # PRIMARY TEST: One-Way ANOVA
> # Assumption check: homogeneity of variance
> leveneTest(payment_value ~ payment_type, data = df_clean)
Levene's Test for Homogeneity of Variance (center = median)
    Df F value    Pr(>F)
group  3  34.547 < 2.2e-16 ***
    97912
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> # Standard ANOVA
> aov_model <- aov(payment_value ~ payment_type, data = df_clean)
> summary(aov_model)
              Df    Sum Sq Mean Sq F value Pr(>F)
payment_type  3  1.129e+07 3762433   78.43 <2e-16 ***
Residuals    97912  4.697e+09  47973
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> # Levene's test is expected to be significant (unequal variances),
> # so Welch's ANOVA is used as the trusted result:
> oneway.test(payment_value ~ payment_type, data = df_clean, var.equal = FALSE)

One-way analysis of means (not assuming equal variances)

data: payment_value and payment_type
F = 95.288, num df = 3.0, denom df = 4876.2, p-value < 2.2e-16
```

Image: Tukey HSD post-hoc pairwise comparisons

```

89 # =====
90 # PRIMARY TEST: One-Way ANOVA
91 # =====
92
93 # Assumption check: homogeneity of variance
94 leveneTest(payment_value ~ payment_type, data = df_clean)
95
96 # Standard ANOVA
97 aov_model <- aov(payment_value ~ payment_type, data = df_clean)
98 summary(aov_model)
99
100 # Levene's test is expected to be significant (unequal variances),
101 # so Welch's ANOVA is used as the trusted result:
102 oneway.test(payment_value ~ payment_type, data = df_clean, var.equal = FALSE)
103
104 # Post-hoc pairwise comparisons
105 TukeyHSD(aov_model)
106
94.48 [Untitled]
R Script

```

Console

```

(R - R 4.6.1 - /cloud/project/...)
> summary(aov_model)
      Df Sum Sq Mean Sq  F value    Pr(>F)
payment_type    3  1.129e+07 3762433  78.43 <2e-16 ***
Residuals  97912  4.697e+09  47973
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> # Levene's test is expected to be significant (unequal variances),
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> oneway.test(payment_value ~ payment_type, data = df_clean, var.equal = FALSE)

One-way analysis of means (not assuming equal variances)

data: payment_value and payment_type
F = 95.288, num df = 3.0, denom df = 4876.2, p-value < 2.2e-16

> # Post-hoc pairwise comparisons
> TukeyHSD(aov_model)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = payment_value ~ payment_type, data = df_clean)

$payment_type
              diff            lwr            upr      p adj
credit_card-boleto    21.794530    17.26444    26.324621 0.0000000
debit_card-boleto     -2.679016   -17.69192    12.333888 0.9679957
voucher-boleto        -19.372555   -31.07987    -7.665243 0.0001251
debit_card-credit_card -24.473546   -39.08132   -9.865771 0.0000987
voucher-credit_card    -41.167085   -52.35015   -29.984020 0.0000000
voucher-debit_card     -16.693539   -34.85742    1.470340 0.0847283

```

```

# Levene's Test for Homogeneity of Variance
      Df      F value Pr(>F)
group   3 34.54733 2.62994e-22
      97912      NA      NA

# ANOVA
      Df      Sum Sq Mean Sq F value Pr(>F)
payment_type    3  1.129e+07 3762433  78.43 <2e-16 ***
Residuals  97912  4.697e+09  47973

# Welch's ANOVA (robust to unequal variance)
One-way analysis of means (not assuming equal variances)
data: payment_value and payment_type
F = 95.288, num df = 3.0, denom df = 4876.2, p-value < 2.2e-16

# Tukey HSD post-hoc comparisons
              diff            lwr            upr      p adj
credit_card-boleto    21.794530    17.26444    26.324621 0.0000000
debit_card-boleto     -2.679016   -17.69192    12.333888 0.9679957
voucher-boleto        -19.372555   -31.07987    -7.665243 0.0001251
debit_card-credit_card -24.473546   -39.08132   -9.865771 0.0000987
voucher-credit_card    -41.167085   -52.35015   -29.984020 0.0000000
voucher-debit_card     -16.693539   -34.85742    1.470340 0.0847283

```

Findings:

The Levene's test signals that the equal-variance assumption was violated ($F = 34.55$, $p < .001$), Hence Welch's ANOVA that does not assume the equal variances, is treated as primary result. Welch's ANOVA presents a highly significant difference ($F = 95.29$, $p < .001$), consistent with the standard ANOVA ($F(3, 97912) = 78.43$, $p < .001$).

The Tukey's post-hoc comparisons presented that credit_card is associated with a higher payment_value in comparison to every other payment type (vs boleto: +\$21.79, $p < .001$; vs debit_card: +\$24.47, $p < .001$; vs voucher: +\$41.17, $p < .001$), and voucher is significantly lower than boleto (-\$19.37, $p < .001$). The payment type, Debit_card did not significantly differ from boleto or voucher.

H0 is rejected since there is statistically significant difference in the mean payment_value with all payment types. It is to be noted that the transactions by credit card are higher in value in comparison to other payment types.

Supporting check: Independent t-test (Credit Card and Boleto)

Focused on the two largest payment categories, two-group validation of the ANOVA finding.

```
t_data <- df_clean %>% filter(payment_type %in% c("credit_card", "boleto"))
t_data$payment_type <- droplevels(t_data$payment_type)

leveneTest(payment_value ~ payment_type, data = t_data)
t.test(payment_value ~ payment_type, data = t_data)
```

Output:

The screenshot shows the RStudio interface with the following content:

- Script Editor:** Contains R code for filtering data by payment type and performing statistical tests.
- Environment:** Lists objects in the global environment, including data frames like `df_clean` and `df_final`.
- Console:** Displays the output of the tests:


```
> # SUPPORTING CHECK A: Independent t-test (credit_card vs boleto)
> #
> t_data <- df_clean %>% filter(payment_type %in% c("credit_card", "boleto"))
> t_data$payment_type <- droplevels(t_data$payment_type)
> leveneTest(payment_value ~ payment_type, data = t_data)
Levene's Test for Homogeneity of Variance (center = median)
  Df F value Pr(>F)
group 1 64.423 1.015e-15 ***
93779
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> t.test(payment_value ~ payment_type, data = t_data)

Welch Two Sample t-test

data: payment_value by payment_type
t = -12.808, df = 32147, p-value < 2.2e-16
alternative hypothesis: true difference in means between group boleto and group credit_card is not equal to 0
95 percent confidence interval:
 -25.12982 -18.45924
sample estimates:
 mean in group boleto mean in group credit_card
      144.3875          166.1820
```

```
# Levene's Test
      Df      F value Pr(>F)
group  1 64.42284 1.015388e-15
93779      NA      NA

# Welch Two Sample t-test
data: payment_value by payment_type
t = -12.808, df = 32147, p-value < 2.2e-16
95 percent confidence interval:
 -25.12982 -18.45924
```

```
sample estimates:
      mean in group boleto  mean in group credit_card
      144.3875              166.1820
```

Findings:

The T-Test results matches with the ANOVA results for specifically this pair, Credit Card and Boleto. Credit Card customers have spent more than boleto customers with credit card at (\$166.18) and Boleto at (\$144.39), $t = -12.808$, $p < .001$, 95% CI: \$18.46 to \$25.13. It matches the ANOVA Tukey comparison for almost the same pair. As expected from these two methods. It tests the same underlying relationship

Supporting check: Simple Linear Regression

The same relationship as a regression mode that is mathematically equivalent to the ANOVA.

```
lm_payment <- lm(payment_value ~ payment_type, data = df_clean)
summary(lm_payment)
```

Output:

The screenshot shows the RStudio interface with the following components:

- Script Editor:** Contains R code for data cleaning, model fitting, and summary generation.
- Environment:** Lists objects in the global environment, including data frames and model objects.
- Files:** Shows the project file structure.
- Console:** Displays the output of the R commands, including the summary of the linear model.

Console Output:

```
> > # =====
> # SUPPORTING CHECK B: Simple linear regression
> # =====
> t_data <- df_clean %>% filter(payment_type %in% c("credit_card", "boleto"))
> t_data$payment_type <- droplevels(t_data$payment_type)
> leveneTest(payment_value ~ payment_type, data = t_data)
> t.test(payment_value ~ payment_type, data = t_data)
> # =====
> # SUPPORTING CHECK B: Simple linear regression
> # =====
> lm_payment <- lm(payment_value ~ payment_type, data = df_clean)
> summary(lm_payment)

Call:
lm(formula = payment_value ~ payment_type, data = df_clean)

Residuals:
    Min       1Q   Median       3Q      Max
-156.6   -98.1   -54.8    17.2  13497.9

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  144.387    1.570   91.987 < 2e-16 ***
payment_typecredit_card  21.795    1.763  12.360 < 2e-16 ***
payment_typedebit_card  -2.679    5.844  -0.458  0.647
payment_typevoucher   -19.373    4.557  -4.251 2.13e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 219 on 97912 degrees of freedom
Multiple R-squared:  0.002397, Adjusted R-squared:  0.002367
F-statistic: 78.43 on 3 and 97912 DF, p-value: < 2.2e-16
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  144.387    1.570   91.987 < 2e-16 ***
payment_typecredit_card  21.795    1.763  12.360 < 2e-16 ***
payment_typedebit_card  -2.679    5.844  -0.458  0.647
payment_typevoucher   -19.373    4.557  -4.251 2.13e-05 ***

Residual standard error: 219 on 97912 degrees of freedom
Multiple R-squared:  0.002397, Adjusted R-squared:  0.002367
F-statistic: 78.43 on 3 and 97912 DF, p-value: < 2.2e-16
```

Findings:

The coefficients of the regressions match with the ANOVA/Tukey results perfectly (credit_card: +\$21.80 vs boleto baseline, $p < .001$; voucher: -\$19.37, $p < .001$). This confirms that the two approaches agree. $R^2 = 0.0024$ signals that payment_type alone explains only 0.24% of the variance in payment_value. The relationship is statistically robust the effect size is relatively small with reference to the total variation in the transaction size. This means that other factors drive most of the variation in what is spent by the customers

Recommendations

As evident from the analysis that credit card usage is consistently linked with higher transactions value in all three tests (ANOVA, T-test, Linear regression). It could be said that the business should incentivize credit card payments specifically in order for a business gain. For instance, an installment promotion could be implemented that would increase the average order value, specifically for the high-priced items. Furthermore, voucher-based transactions are with the lowest value that indicates the purchases were mainly promotional or discount driven. Voucher purchased can not be termed a segment to be targeted for value growth.

Furthermore, given that effect is concentrated with the credit cards installment promotion instead of the card usage, the next step that could be tested is whether the number of installment adjustment offered at the checkout brings meaningful value as an increase in average order value or not. This could be further explored with a subset of the products categories that would help the business to check the mechanism suggested by this analysis. However, it's not recommended to be rolled out platform wide before there is concrete evidence to support the claim that whether installment adjustment at the checkout brings meaningful value. As the effect size found here is relatively quite small in absolute terms ($R^2 = 0.0024$) and such intervention should be evaluated primarily with the marginal impact on the average order value, instead of expecting to bring a major shift for individual transactions.

Limitations

Since this is observational and cross-sectional data, the relationship cannot be interpreted casually. The payment method choice could be confounded with other unmeasured characteristics and factors that aren't taken into account. For instance, purchase intention or income which directly independently influences the payment method and total spending. The R^2 confirms this that payment type is one of the many factors which influence the total transactional value and not a single dominant one. Moreover, by aggregating multi-items and installment orders to one row per order requires narrowing down the assumptions. In particular, taking the first recorded payment type per order could lead to misrepresenting the small number order that were paid with more than one payment method.

Future Work

A future work could be a multiple regression incorporating additional the predictors driving the transactional value such as the product category, delivery time, region and alongside the payment methods which could clarify that how much of this effect is independent of other purchasing factors.

GitHub Repository

<https://github.com/Awad265/B105-Individual-Project.git>

References

Kaggle (2021). *Brazilian e-commerce public dataset by olist*. [online] www.kaggle.com. Available at: <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce> [Accessed 16 Sept. 2026].